

2026.6.9. Fourier 级数.

0. 常用积分回顾.

$$\textcircled{1} \int_0^\varphi \sin nx \, dx \xrightarrow{t=nx} \frac{1}{n} \int_0^{n\varphi} \sin t \, dt = \frac{1}{n} \cos t \Big|_{n\varphi}^0 \\ = \frac{1}{n} (1 - \cos n\varphi)$$

$$\int_0^\varphi \cos nx \, dx \xrightarrow{t=nx} \frac{1}{n} \int_0^{n\varphi} \cos t \, dt = \frac{\sin t}{n} \Big|_0^{n\varphi} = \frac{\sin n\varphi}{n}$$

$$\textcircled{2} \int_0^\pi x \sin nx \, dx \xrightarrow{t=nx} \frac{1}{n^2} \int_0^{n\pi} t \sin t \, dt.$$

$$\int t \sin t \, dt = - \int t \, d\cos t = -t \cos t + \int \cos t \, dt \\ = \sin t - t \cos t.$$

$$\text{原积分} = \frac{1}{n^2} (\sin t - t \cos t) \Big|_0^{n\pi} = \frac{1}{n^2} [(0 - (-1)^n n\pi) - 0] \\ = \frac{(-1)^{n+1} n\pi}{n^2} = \frac{(-1)^{n+1} \pi}{n}.$$

$$\int_0^\pi x \cos nx \, dx \xrightarrow{t=nx} \frac{1}{n^2} \int_0^{n\pi} t \cos t \, dt$$

$$\int t \cos t \, dt = \int t \, d\sin t = t \sin t - \int \sin t \, dt = t \sin t + \cos t.$$

$$\text{原积分} = \frac{1}{n^2} (t \sin t + \cos t) \Big|_0^{n\pi} = \frac{1}{n^2} [(-1)^n - 1] = \begin{cases} 0, & 2|n \\ -\frac{2}{n^2}, & 2 \nmid n. \end{cases}$$

$$\textcircled{3} \int_0^\pi x^2 \sin nx \, dx \xrightarrow{t=nx} \frac{1}{n^3} \int_0^{n\pi} t^2 \sin t \, dt.$$

$$\int t^2 \sin t \, dt = - \int t^2 \, d\cos t = -t^2 \cos t + \int 2t \cos t \, dt \\ = -t^2 \cos t + 2 \int t \, d\sin t = -t^2 \cos t + 2t \sin t - 2 \int \sin t \, dt \\ = -t^2 \cos t + 2t \sin t + 2 \cos t.$$

$$\text{原积分} = \frac{1}{n^3} (-t^2 \cos t + 2t \sin t + 2 \cos t) \Big|_0^{n\pi} \\ = \frac{1}{n^3} [(-n^2 \pi^2 \cdot (-1)^n + 0 + 2(-1)^n) - (0 + 0 + 2)] \\ = \frac{(-1)^{n+1} \pi^2}{n} + \frac{2}{n^3} [(-1)^n - 1].$$

$$\int_0^{\pi} x^2 \cos nx dx \stackrel{t=nx}{=} \frac{1}{n^3} \int_0^{n\pi} t^2 \cos t dt.$$

$$\begin{aligned} \int t^2 \cos t dt &= \int t^2 d \sin t = t^2 \sin t - \int \sin t \cdot 2t dt \\ &= t^2 \sin t + 2 \int t d \cos t = t^2 \sin t + 2t \cos t - 2 \int \cos t dt \\ &= t^2 \sin t + 2t \cos t - 2 \sin t \end{aligned}$$

$$\begin{aligned} \text{原积分} &= \frac{1}{n^3} \left[(t^2 \sin t + 2t \cos t - 2 \sin t) \Big|_0^{n\pi} \right] \\ &= \frac{1}{n^3} \left[(0 + 2n\pi(-1)^n - 0) - (0 + 0 - 0) \right] \\ &= \frac{2\pi}{n^2} (-1)^n. \end{aligned}$$

④ $\int e^{\alpha x} \sin \beta x dx, \int e^{\alpha x} \cos \beta x dx.$

注意到 $(e^{\alpha x} \sin \beta x)' = e^{\alpha x} (\alpha \sin \beta x + \beta \cos \beta x)$

$$(e^{\alpha x} \cos \beta x)' = e^{\alpha x} (-\beta \sin \beta x + \alpha \cos \beta x).$$

$$\begin{aligned} \text{那么, } (A e^{\alpha x} \sin \beta x + B e^{\alpha x} \cos \beta x)' \\ = e^{\alpha x} [(A\alpha - B\beta) \sin \beta x + (A\beta + B\alpha) \cos \beta x]. \end{aligned}$$

$$(i) \quad \begin{cases} \alpha \cdot A - \beta \cdot B = 1 \\ \beta \cdot A + \alpha \cdot B = 0. \end{cases} \quad \text{即} \quad \begin{cases} A = \frac{\alpha}{\alpha^2 + \beta^2} \\ B = \frac{-\beta}{\alpha^2 + \beta^2}. \end{cases}$$

$$\text{即} \int e^{\alpha x} \sin \beta x dx = \frac{e^{\alpha x}}{\alpha^2 + \beta^2} (\alpha \sin \beta x - \beta \cos \beta x)$$

$$(ii) \quad \begin{cases} \alpha \cdot A - \beta \cdot B = 0 \\ \beta \cdot A + \alpha \cdot B = 1. \end{cases} \quad \text{即} \quad \begin{cases} A = \frac{\beta}{\alpha^2 + \beta^2} \\ B = \frac{\alpha}{\alpha^2 + \beta^2} \end{cases}$$

$$\text{即} \int e^{\alpha x} \cos \beta x dx = \frac{e^{\alpha x}}{\alpha^2 + \beta^2} (\beta \sin \beta x + \alpha \cos \beta x).$$

「关于 Fourier 级数的和函数」.

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx) \triangleq S(x)$$

① Dirichlet 定理.

$f(x)$ 在 $[-\pi, \pi]$ 上分段连续、分段单调.

$$\text{则 } S(x) = \begin{cases} f(x), & x \text{ 是连续点,} \\ \frac{1}{2} (f(x^+) + f(x^-)), & x \text{ 是间断点.} \end{cases}$$

② $f(x)$ 在 $[-\pi, \pi]$ 上分段可微, 收敛性同上.

Rmk. 根据以上判断 $S(x)$, 并不需求出 a_n, b_n .

1. 求 Fourier 系数.

① $f(x) = x, \quad x \in [-\pi, \pi].$

由奇性, $a_n = 0$.

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx \, dx = \frac{2(-1)^{n+1}}{n}.$$

$$\Rightarrow x \sim 2 \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{n} \sin nx = \begin{cases} x, & x \in (-\pi, \pi) \\ 0, & x = \pm \pi \end{cases}$$

② $f(x) = |x|, \quad x \in [-\pi, \pi].$

由偶性, $b_n = 0$.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \, dx = \pi.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx = \begin{cases} 0, & 2 \mid n \\ -\frac{4}{n^2 \pi}, & 2 \nmid n. \end{cases}$$

$$\Rightarrow |x| \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{+\infty} \frac{\cos(2n-1)x}{(2n-1)^2} = |x|, \quad x \in [-\pi, \pi].$$

$$\text{令 } x = \pi. \Rightarrow \pi = \frac{\pi}{2} + \frac{4}{\pi} \sum_{n=1}^{+\infty} \frac{1}{(2n-1)^2} \Rightarrow \sum_{n=1}^{+\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

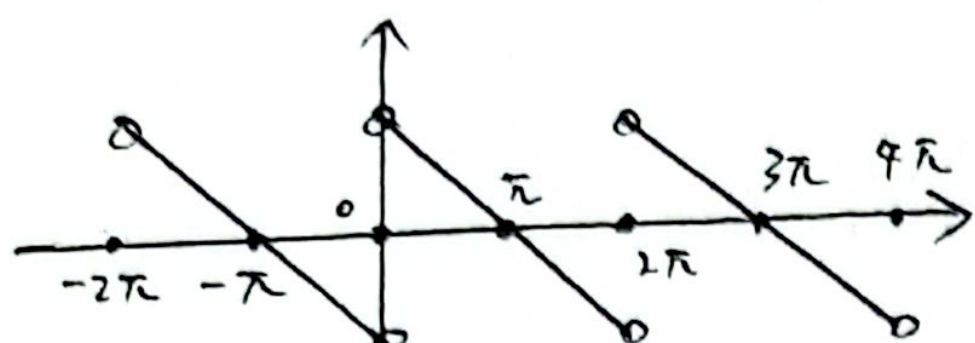
Rmk. 以上两例也可看作 $f(x) = x, \quad x \in [0, \pi]$ 的

奇延拓/正弦展开 和 偶延拓/余弦展开.

$$\Rightarrow \frac{\pi}{2} = \frac{4}{\pi} \sum_{n=1}^{+\infty} \frac{\cos(2n-1)x}{(2n-1)^2} + 2 \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{n} \sin nx, \quad x \in [0, \pi).$$

③ $f(x) = \frac{\pi-x}{2}, x \in [0, 2\pi]$.

这也可看作 $f(x) = \frac{\pi-x}{2}, x \in [0, \pi]$ 做奇延拓



故 $a_n = 0$.

$$b_n = \frac{2}{\pi} \int_0^{\pi} \frac{\pi-x}{2} \sin nx \, dx = \int_0^{\pi} \sin nx \, dx - \frac{1}{\pi} \int_0^{\pi} x \sin nx \, dx$$

$$= \frac{1}{n} (1 - (-1)^n) - \frac{1}{\pi} \cdot \frac{(-1)^{n+1}}{n} \pi = \frac{1}{n} - \frac{(-1)^n}{n} + \frac{(-1)^n}{n} = \frac{1}{n}.$$

$$\Rightarrow \frac{\pi-x}{2} \sim \sum_{n=1}^{+\infty} \frac{\sin nx}{n} = \begin{cases} \frac{\pi-x}{2}, & x \in (0, 2\pi) \\ 0, & x = 0, 2\pi. \end{cases}$$

这个可以记住!

Rmk. 答案并不是 $\frac{\pi}{2}$ 或去 ①② 中任何一个.

不要加减 Fourier 级数. 每个都好好算!

④ $f(x) = \frac{x^2}{4} - \frac{\pi}{2}x, x \in [0, 2\pi]$.

注意到 $f(0) = f(2\pi)$. 以及对称轴 $x = \pi$.

这可看作 $f(x) = \frac{x^2}{4} - \frac{\pi}{2}x, x \in [0, \pi]$ 做偶延拓.

$$b_n = 0, \quad a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = -\frac{\pi^2}{3}.$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx = \frac{1}{2\pi} \int_0^{\pi} x^2 \cos nx \, dx - \int_0^{\pi} x \cos nx \, dx$$

$$= \frac{1}{2\pi} \cdot \frac{2\pi}{n^2} (-1)^n - \frac{1}{n^2} ((-1)^n - 1) = \frac{1}{n^2}.$$

$$\Rightarrow \frac{x^2}{4} - \frac{\pi}{2}x = -\frac{\pi^2}{6} + \sum_{n=1}^{+\infty} \frac{\cos nx}{n^2}, \quad x \in [0, 2\pi]$$

令 $x=0$. 得 $\sum_{n=1}^{+\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$

例 2. $\frac{\pi-x}{2} = \sum_{n=1}^{+\infty} \frac{\sin nx}{n}$, $x \in (-\pi, \pi)$ 内一致收敛.

因此可以逐项积分.

$$\Rightarrow \frac{\pi}{2}x - \frac{x^2}{4} = C + \sum_{n=1}^{+\infty} \frac{\cos nx}{n^2}$$

$$\Rightarrow \frac{x^2}{4} - \frac{\pi}{2}x = \frac{a_0}{2} + \sum_{n=1}^{+\infty} \frac{\cos nx}{n^2} = -\frac{\pi^2}{6} + \sum_{n=1}^{+\infty} \frac{\cos nx}{n^2}$$

⑤ $f(x) = \begin{cases} 1, & |x| \leq \varphi \\ 0, & |x| > \varphi \end{cases} \quad \varphi \in (0, \pi).$

$b_n = 0$ (偶函数).

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2\varphi}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\varphi} \cos nx dx = \frac{2 \sin n\varphi}{n\pi}$$

$$\Rightarrow f(x) \sim \frac{\varphi}{\pi} + \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n} \cos nx$$

$$= \begin{cases} 1, & |x| < \varphi \\ \frac{1}{2}, & |x| = \varphi \\ 0, & \varphi < |x| \leq \pi \end{cases}$$

令 $x=0$. 得 $1 = \frac{\varphi}{\pi} + \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n}$

$$\text{即 } \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n} = \frac{\pi}{2} \cdot \left(1 - \frac{\varphi}{\pi}\right) = \frac{\pi - \varphi}{2}, \quad \varphi \in (0, \pi).$$

⑥ $f(x) = \begin{cases} 1, & 0 \leq x \leq \varphi \\ -1, & -\varphi \leq x < 0 \\ 0, & \varphi < |x| \leq \pi \end{cases} \quad a_n = 0$ (奇函数).

$$b_n = \frac{2}{\pi} \int_0^{\varphi} \sin nx dx = \frac{2}{\pi} \cdot \frac{1}{n} (1 - \cos n\varphi)$$

$$\Rightarrow f(x) \sim \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{1 - \cos n\varphi}{n} \sin nx = \begin{cases} 1, & 0 < x \leq \varphi \\ -1, & -\varphi < x < 0 \\ 0, & x=0 \text{ 或 } \varphi < |x| \leq \pi \\ \frac{1}{2}, & x = \varphi \\ -\frac{1}{2}, & x = -\varphi \end{cases}$$

⑦ $f(x) = e^x, \quad x \in [-\pi, \pi].$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x dx = \frac{1}{\pi} (e^{\pi} - e^{-\pi})$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \cos nx dx = \frac{1}{\pi} \cdot \frac{1}{1+n^2} e^x (n \sin nx + \cos nx) \Big|_{-\pi}^{\pi} = \frac{(-1)^n}{\pi(1+n^2)} (e^{\pi} - e^{-\pi})$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin nx dx = \frac{1}{\pi} \cdot \frac{1}{1+n^2} e^x (\sin nx - n \cos nx) \Big|_{-\pi}^{\pi} = \frac{(-1)^{n+1} n}{\pi(1+n^2)} (e^{\pi} - e^{-\pi})$$

$$\Rightarrow e^x \sim \frac{e^{\pi} - e^{-\pi}}{2\pi} + \frac{e^{\pi} - e^{-\pi}}{\pi} \sum_{n=1}^{+\infty} \frac{(-1)^n}{n^2+1} (\cos nx - \sin nx) = \begin{cases} e^x, & x \in (-\pi, \pi) \\ \frac{e^{\pi} + e^{-\pi}}{2}, & x = \pm \pi \end{cases}$$

令 $x = \pi$
 $\Rightarrow \sum_{n=1}^{+\infty} \frac{1}{n^2+1} = \pi \frac{e^{\pi} + e^{-\pi}}{e^{\pi} - e^{-\pi}} - \frac{1}{2}$

2. Parseval 等式.

如果 $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx)$, $x \in [-\pi, \pi]$.

$$\text{则 } \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx = \frac{a_0^2}{2} + \sum_{n=1}^{+\infty} (a_n^2 + b_n^2).$$

理解: Fourier 展开是函数 f 到序列 $\{a_0, a_1, b_1, a_2, b_2, \dots\}$ 的映射. 映射前后, 对象的“长度”一样. 即 Fourier 展开是保距映射.

一般考察方式为给定 $f(x)$, 求 Fourier 展开, 再用 Parseval 等式求 $\sum_{n=1}^{+\infty} (a_n^2 + b_n^2) + \frac{a_0^2}{2}$ 的值.

$$\textcircled{1} \sum_{n=1}^{+\infty} \frac{\sin nx}{n} \quad \text{则 } \sum_{n=1}^{+\infty} \frac{1}{n^2} = \frac{1}{\pi} \int_{-\pi}^{\pi} \left(\frac{\pi-x}{2} \right)^2 dx = \frac{\pi^2}{6}$$

$\frac{\pi-x}{2} \sim$

$$\textcircled{2} \frac{x^2}{4} - \frac{\pi}{2}x \sim -\frac{\pi^2}{6} + \sum_{n=1}^{+\infty} \frac{\cos nx}{n^2}.$$

$a_0 = -\frac{\pi^2}{3}$

$$\text{则 } \sum_{n=1}^{+\infty} \frac{1}{n^4} + \frac{1}{2} \times \frac{\pi^4}{9} = \frac{1}{\pi} \int_{-\pi}^{\pi} \left(\frac{x^2}{4} - \frac{\pi}{2}x \right)^2 dx = \frac{\pi^4}{15}.$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{1}{n^4} = \frac{\pi^4}{15} - \frac{\pi^4}{18} = \frac{\pi^4}{90}.$$

$$\textcircled{3} f(x) = \begin{cases} 1, & |x| \leq \varphi \\ 0, & \varphi < |x| \leq \pi \end{cases} \sim \frac{\varphi}{\pi} + \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n} \cdot \cos nx. \quad \text{求 } \sum_{n=1}^{+\infty} \frac{\cos n\varphi}{n^2}$$

$$\Rightarrow \frac{1}{\pi} \int_{-\varphi}^{\varphi} dx = \frac{1}{2} \cdot \left(\frac{2\varphi}{\pi} \right)^2 + \frac{4}{\pi^2} \sum_{n=1}^{+\infty} \frac{\sin^2 n\varphi}{n^2}$$

$\frac{2\varphi}{\pi}$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{\sin^2 n\varphi}{n^2} = \frac{\pi^2}{4} \cdot \left(\frac{2\varphi}{\pi} - \frac{2\varphi^2}{\pi^2} \right) = \frac{\pi\varphi}{2} - \frac{\varphi^2}{2}.$$

$$\text{又 } \sin^2 n\varphi = \frac{1 - \cos 2n\varphi}{2}$$

$$\text{故 } \frac{1}{2} \sum_{n=1}^{+\infty} \frac{1}{n^2} - \frac{1}{2} \sum_{n=1}^{+\infty} \frac{\cos 2n\varphi}{n^2} = \frac{\pi\varphi}{2} - \frac{\varphi^2}{2}$$

$$\text{故 } \sum_{n=1}^{+\infty} \frac{\cos 2n\varphi}{n^2} = \varphi^2 - \pi\varphi + \frac{\pi^2}{6}$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{\cos n\varphi}{n^2} = \left(\frac{\varphi}{2}\right)^2 - \pi \cdot \frac{\varphi}{2} + \frac{\pi^2}{6} = \frac{\varphi^2}{4} - \frac{\pi\varphi}{2} + \frac{\pi^2}{6}.$$

④ 求 $\sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n^3}$

「内积形式 Parseval 等式」.

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx)$$

$$g(x) \sim \frac{c_0}{2} + \sum_{n=1}^{+\infty} (c_n \cos nx + d_n \sin nx)$$

$$\text{则 } \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)g(x)dx = \frac{a_0c_0}{2} + \sum_{n=1}^{+\infty} (a_nc_n + b_nd_n)$$

$$\text{证明: } f(x) \pm g(x) \sim \frac{a_0 \pm c_0}{2} + \sum_{n=1}^{+\infty} ((a_n \pm c_n) \cos nx + (b_n \pm d_n) \sin nx).$$

$$\Rightarrow \frac{1}{\pi} \int_{-\pi}^{\pi} (f+g)^2 dx = \frac{(a_0+c_0)^2}{2} + \sum_{n=1}^{+\infty} ((a_n+c_n)^2 + (b_n+d_n)^2)$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} (f-g)^2 dx = \frac{(a_0-c_0)^2}{2} + \sum_{n=1}^{+\infty} ((a_n-c_n)^2 + (b_n-d_n)^2).$$

作差即得. $\square$.

$$\text{用 } \frac{x^2}{4} - \frac{\pi}{2}x \sim -\frac{\pi^2}{6} + \sum_{n=1}^{+\infty} \frac{\cos nx}{n^2}$$

$$\begin{cases} 1, & |x| \leq \varphi \\ 0, & \varphi < |x| \leq \pi \end{cases} \sim \frac{\varphi}{\pi} + \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n} \cdot \cos nx$$

$$\text{有 } \frac{2}{\pi} \int_0^{\varphi} \left(\frac{x^2}{4} - \frac{\pi}{2}x\right) dx = \frac{1}{2} \cdot \left(-\frac{\pi^2}{3}\right) \cdot \frac{2\varphi}{\pi} + \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n^3}$$

$$\text{即 } \frac{\varphi^3}{6\pi} - \frac{\varphi^2}{2} = -\frac{\pi\varphi}{3} + \frac{2}{\pi} \sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n^3}$$

$$\sum_{n=1}^{+\infty} \frac{\sin n\varphi}{n^3} = \frac{\varphi^3}{12} - \frac{\varphi^2\pi}{4} + \frac{\varphi\pi^2}{6}, \quad \varphi \in (0, \pi), \quad (\text{对 } \varphi \in [0, \pi] \text{ 实际也对}).$$

3. Fourier 级数的一致收敛性.

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx).$$

「定理」. $f(x)$ 以 2π 为周期, $f'(x) \in R[-\pi, \pi]$.

则 Fourier 级数在 $\mathbb{R}$ 上 $\Rightarrow f(x)$.

证明: 设 $f'(x) \sim \frac{a'_0}{2} + \sum_{n=1}^{+\infty} (a'_n \cos nx + b'_n \sin nx)$

$$a'_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f'(x) dx = \frac{1}{\pi} (f(\pi) - f(-\pi)) = 0.$$

$$\begin{aligned} a'_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f'(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos nx df(x) \\ &= \frac{1}{\pi} \underbrace{f(x) \cos nx}_{=0} \Big|_{-\pi}^{\pi} - \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) d \cos nx \end{aligned}$$

$$= -\frac{n}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = n b_n.$$

$$\begin{aligned} b'_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f'(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin nx df(x) \\ &= \frac{1}{\pi} \underbrace{f(x) \sin nx}_{=0} \Big|_{-\pi}^{\pi} - \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) d \sin nx \\ &= -\frac{n}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = -n a_n. \end{aligned}$$

$$\text{故 } f'(x) \sim \sum_{n=1}^{+\infty} (n b_n \cos nx - n a_n \sin nx).$$

Rmk. 这正是 $\frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx)$ 逐项微分!

由 Parseval 等式.

$$\begin{aligned} \sum_{n=1}^{+\infty} (|a_n| + |b_n|) &= \sum_{n=1}^{+\infty} \frac{1}{n} (|a'_n| + |b'_n|) \\ &\leq \sum_{n=1}^{+\infty} \frac{1}{n^2} \cdot \sum_{n=1}^{+\infty} ((a'_n)^2 + (b'_n)^2) \\ &= \frac{\pi^2}{6} \cdot \frac{1}{\pi} \int_{-\pi}^{\pi} (f'(x))^2 dx < +\infty. \end{aligned}$$

由 M-判别法, 知 $\frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx) \Rightarrow f(x). \square$

4. Fourier 级数的唯一性.

$f(x)$ 和 $g(x)$ 对应相同 Fourier 级数 $\Leftrightarrow \int_{-\pi}^{\pi} |f(x) - g(x)|^2 dx = 0$.

证明. 设 $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx)$

$$g(x) \sim \frac{c_0}{2} + \sum_{n=1}^{+\infty} (c_n \cos nx + d_n \sin nx)$$

$$f(x) - g(x) \sim \frac{a_0 - c_0}{2} + \sum_{n=1}^{+\infty} ((a_n - c_n) \cos nx + (b_n - d_n) \sin nx)$$

$$\text{则 } \frac{1}{\pi} \int_{-\pi}^{\pi} |f(x) - g(x)|^2 dx = \frac{1}{2} (a_0 - c_0)^2 + \sum_{n=1}^{+\infty} ((a_n - c_n)^2 + (b_n - d_n)^2)$$

$$\text{LHS} = 0 \Leftrightarrow \text{RHS} = 0 \Leftrightarrow a_n = c_n, b_n = d_n, \forall n. \quad \square$$

~~Rmk.~~ Taylor 级数不具有这种唯一性.

例如 $f(x) \sim \sum_{n=1}^{+\infty} \frac{f^{(n)}(0)}{n!} x^n$.

由于 $\begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ 在 0 处任意阶导数为 0.

故 $f(x) + e^{-\frac{1}{x^2}} \sim \sum_{n=1}^{+\infty} \frac{f^{(n)}(0)}{n!} x^n$.

但 $|f(x) + e^{-\frac{1}{x^2}} - f(x)| = |e^{-\frac{1}{x^2}}| \neq 0$.

5. Poincaré 不等式.

$f \in C'[0, 2\pi], f(0) = f(2\pi), \int_0^{2\pi} f(x) dx = 0$. 则 $\int_0^{2\pi} (f'(x))^2 dx \geq \int_0^{2\pi} f(x)^2 dx$

证明. $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx)$ $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = 0$.

$$f'(x) \sim \sum_{n=1}^{+\infty} (n b_n \cos nx - n a_n \sin nx)$$

$$\text{LHS} = \pi \cdot \sum_{n=1}^{+\infty} (n^2 b_n^2 + n^2 a_n^2) \geq \pi \cdot \sum_{n=1}^{+\infty} (a_n^2 + b_n^2) = \text{RHS}. \quad \square$$

Rmk. 由证明还知道取等号 $\Leftrightarrow f(x) = A \sin x + B \cos x$.

一个有趣的应用是等周不等式. L 是简单闭曲线的长度, A 为内部面积

则 $L^2 \geq 4\pi A$. 取等 $\Leftrightarrow C$ 是圆. t 为弧长, $L=2\pi$ 那么 $(x'(t))^2 + (y'(t))^2 = \frac{dx^2 + dy^2}{dt^2} = 1$.

证明: 参数化 $C \begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad t \in [0, 2\pi], \quad x(0) = x(2\pi), \quad y(0) = y(2\pi)$
并平移使 $\int_0^{2\pi} x(t) dt = \int_0^{2\pi} y(t) dt = 0$.

那么, $L = \int_0^{2\pi} \sqrt{(x'(t))^2 + (y'(t))^2} dt$

$$\Rightarrow A \leq \pi = \frac{L^2}{4\pi} \quad \square$$

$$4A^2 = \left(\int_0^{2\pi} (xy' - x'y) dt \right)^2 \leq \int_0^{2\pi} x^2 + y^2 dt \int_0^{2\pi} (x')^2 + (y')^2 dt \leq \left(\int_0^{2\pi} (x')^2 + (y')^2 dt \right)^2 = 4\pi^2$$

6. 一道往年问题.

① 求 $\cos \alpha x$, $\alpha \in (0, 1)$ Fourier 展开. $x \in [-\pi, \pi]$.

$$b_n = 0.$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos \alpha x dx = \frac{1}{\alpha \pi} \sin \alpha x \Big|_{-\pi}^{\pi} = \frac{2 \sin \alpha \pi}{\alpha \pi}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos \alpha x \cos nx dx = \frac{2}{\pi} \int_0^{\pi} \cos \alpha x \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \cos(\alpha+n)x + \cos(\alpha-n)x dx$$

$$= \frac{1}{\pi} \left(\frac{\sin(\alpha+n)\pi}{\alpha+n} + \frac{\sin(\alpha-n)\pi}{\alpha-n} \right).$$

$$\begin{aligned} \text{故 } \cos \alpha x &\sim \frac{\sin \alpha \pi}{\alpha \pi} + \frac{1}{\pi} \sum_{n=1}^{+\infty} \left(\frac{\sin(\alpha+n)\pi}{\alpha+n} + \frac{\sin(\alpha-n)\pi}{\alpha-n} \right) \cos nx \\ &= \cos \alpha x, \quad x \in [-\pi, \pi] \quad (\text{因为 } \cos \alpha \pi = \cos(\alpha - (-\pi))) \end{aligned}$$

② 求证: $\frac{\pi}{\sin \alpha \pi} = \frac{1}{\alpha} + \sum_{n=1}^{+\infty} (-1)^n \frac{2\alpha}{\alpha^2 - n^2}.$

对上式中取 $x=0$. ~~由傅里叶级数收敛性~~ $1 = \frac{\sin \alpha \pi}{\alpha \pi} + \frac{1}{\pi} \sum_{n=1}^{+\infty} (-1)^n \left(\frac{\sin \alpha \pi}{\alpha+n} + \frac{\sin \alpha \pi}{\alpha-n} \right)$

变形: $\frac{\pi}{\sin \alpha \pi} = \frac{1}{\alpha} + \sum_{n=1}^{+\infty} (-1)^n \frac{2\alpha}{\alpha^2 - n^2} \quad \square.$

③ 求证: $\frac{\pi}{\sin x} = \frac{\pi}{x} + \sum_{n=1}^{+\infty} (-1)^n \frac{2\pi x}{x^2 - n^2 \pi^2}.$

证明: 令 $\alpha = \frac{x}{\pi}$ 即得 $\square$.

④ 求 $\int_0^{+\infty} \frac{\sin x}{x} dx$.

解: $\frac{1}{\sin x} = \frac{1}{x} + \sum_{n=1}^{+\infty} (-1)^n \left(\frac{1}{x-n\pi} + \frac{1}{x+n\pi} \right).$

$$\frac{\sin x}{x} = 1 - \sum_{n=1}^{+\infty} (-1)^n \left(\frac{\sin x}{x-n\pi} + \frac{\sin x}{x+n\pi} \right)$$

$$= 1 - \sum_{n=1}^{+\infty} \left(\frac{\sin(x-n\pi)}{x-n\pi} + \frac{\sin(x+n\pi)}{x+n\pi} \right).$$

即 $1 = \sum_{n=-\infty}^{+\infty} \frac{\sin(x+n\pi)}{x+n\pi}$ (Dirichlet 判别法一致收敛)

故 $\pi = \int_0^{\pi} dx = \sum_{n=-\infty}^{+\infty} \int_0^{\pi} \frac{\sin(x+n\pi)}{x+n\pi} dx = \sum_{n=-\infty}^{+\infty} \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx = \int_{-\infty}^{+\infty} \frac{\sin x}{x} dx$

由偶性, $\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$

7. 逼近定理介绍.

Parseval 等式说明 $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx) \triangleq S(x)$

$S(x)$ 在 L^2 意义下收敛于 $f(x)$. 即 $\frac{1}{\pi} \int_{-\pi}^{\pi} \left[f(x) - \sum_{n=1}^N (a_n \cos nx + b_n \sin nx) - \frac{a_0}{2} \right]^2 dx \rightarrow 0$.

那么是否可以用三角多项式做到一致逼近 $f(x)$ 呢?

「第二逼近定理」. $f(x) \in C[0, 2\pi]$. 以 2π 为周期. 则存在三角多项式序列 $T_n(x)$ s.t. $T_n(x) \rightarrow f(x)$. $x \in [0, 2\pi]$.

但 $T_n(x)$ 不是 $S_n(x) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx)$.

特别地, 可取 $T_n(x) = \frac{1}{n} (S_1(x) + S_2(x) + \dots + S_n(x))$ 能够满足要求.

「第一逼近定理」. $f(x) \in C[a, b]$. 则存在多项式序列 $P_n(x)$ s.t. $P_n(x) \rightarrow f(x)$.

同样地, $P_n(x)$ 不是 Taylor 多项式.

一般可以对 $f(x) \in C[0, 1]$ 取 $P_n(x) = \sum_{k=0}^n C_n^k f(\frac{k}{n}) x^k (1-x)^{n-k}$ 可满足要求. 称为 Bernstein 多项式.

(*) 高数中, 我们遇到三种逼近, 它们意义不同.

① $f(x) \in C^n(x_0)$. 则在 x_0 处对 f 做 Taylor 展开, $T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$.

有 $f(x) - T_n(x) = o(x^n)$. ($x \rightarrow x_0$). 这是局部最佳逼近.

② $f(x) \in C[a, b]$. 找一列 $f_n(x)$ s.t. $f_n \rightarrow f$. $x \in [a, b]$.

这是一致逼近, 是在距离 $\|f-g\| = \max_{a \leq x \leq b} |f(x) - g(x)|$ 的意义下逼近.

③ $f(x) \in C[0, 2\pi]$. Fourier 展开 $S_n(x) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx)$

有 $\lim_{n \rightarrow \infty} \frac{1}{\pi} \int_0^{2\pi} (f(x) - S_n(x))^2 dx = 0$. 这是 L^2 逼近, 是在

距离 $\int_0^{2\pi} |f(x) - g(x)|^2 dx$ 的意义下逼近.